

Advancements in estimating differential pistons for the Extremely Large Telescope using deep learning

II. Exploiting defocused images

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ABSTRACT

Differential piston aberrations, commonly known as petaling, represent a critical limitation for the optical performance of the Extremely Large Telescopes (ELT). If left uncorrected, petaling severely degrades the core of the point spread function (PSF), restricting high-contrast imaging capabilities and reducing the spatial resolution of a segmented primary mirror to that of a single sub-aperture petal. In this paper, we present an evolution of our deep-learning framework dedicated to estimating differential piston modes. By replacing our previous sub-PSF approach with a method exploiting defocused dual-wavelength focal-plane images, we successfully enhance both the capture range performance and the overall versatility of the wavefront sensor. First, the network can concurrently estimate all five differential piston modes of a 6-petal pupil from a single image frame, introducing native robustness against pupil rotation. Under realistic ELT turbulence residuals (25% Strehl ratio baseline with 13-frame averaging), the network achieves a wavefront error of approximately 25 nm RMS, demonstrating very close agreement with historical frameworks (23 nm RMSE). Second, by incorporating a dual-wavelength input configuration ($\lambda_1 = 1.5 \mu\text{m}$ and $\lambda_2 = 1.8 \mu\text{m}$), the network expands its operational bounds well beyond the classical $[-\pi, \pi]$ phase ambiguity. Robust linear estimation is maintained up to a range of $[-5\pi, 5\pi]$. Beyond the synthetic wavelength limit ($\Lambda = 9.0 \mu\text{m}$), the system continues to exhibit a coarse linear behavior, with no clear evidence of classical phase wrapping. Finally, we outline our roadmap for experimental validation on the High-Contrast Module (HCM) testbed at IPAG using the Zernike wavefront sensor as a reference baseline, paving the way for eventual on-sky deployment on the EKARUS adaptive optics R&D platform at the Asiago Observatory.

Keywords: Petaling, Differential piston, ELT, PSF, Deep Learning

1. INTRODUCTION

The Extremely Large Telescopes is getting close to first light, enabling unprecedented angular resolution and light-gathering power to observe the most interesting and fundamental problems of the Universe. However, differential piston aberrations, commonly known as petaling, can cost up to 60% of the telescope's nominal spatial resolution if left uncorrected. This degradation effectively reduces the ELT to a massive light-collecting bucket with the spatial resolution of only a single one of its six individual petals. Consequently, uncorrected differential piston severely degrades the point spread function (PSF), critically limiting the performance of highly demanding techniques such as high-contrast imaging for exoplanet characterization.

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To sense these phase discontinuities, recent machine-learning frameworks have successfully utilized deep neural networks trained on focal plane data.¹⁻³ Our previously proposed solution relies on the analysis of monochromatic sub-PSFs extracted from a 2×2 Shack-Hartmann wavefront sensor (WFS), extracting focal features to reconstruct individual petal modes. While effective, this sub-PSF approach suffers from two primary bottlenecks: (1) a vulnerability to pupil rotation, and (2) the operational requirement to train two distinct networks to accommodate the two types of subapertures (i.e., those sampling a single differential piston versus those sampling two, see Ref. 3,4). Furthermore, such traditional monochromatic configurations inherently suffer from a restricted capture range, fundamentally bounded by the classical $[-\pi, \pi]$ radian phase ambiguity.

In this work, we propose an evolution of the optical setup established in Ref. 3,4 to overcome these limitations. By adding a calibrated defocus to break the phase even-mode degeneracy⁵ and by using a two-channel monochromatic imaging setup (spanning the H-band at $\lambda_1 = 1.5 \mu\text{m}$ and $\lambda_2 = 1.8 \mu\text{m}$), our approach achieves two significant milestones:

- **Simultaneous multi-petal regression:** The network simultaneously maps a single focal image to all five differential pistons of a 6-petal ELT-like pupil, matching the $\sim 25 \text{ nm}$ RMS accuracy of previous single-piston frameworks while introducing native robustness against pupil rotation.
- **Capture range extension:** By utilizing dual-wavelength diversity, we demonstrate that the network successfully breaks the 2π coherence length barrier, dynamically extending the tolerable piston capture range from the classical $[-\pi, \pi]$ domain up to an range of $[-5\pi, 5\pi]$.

This paper is structured as follows. In Sec. 2, we outline the simulation framework, the properties of the ELT pupil, and the architecture of the ResNet pipeline. Sec. 3 details our numerical results, analyzing prediction accuracy within the classical capture range, benchmarking performance against the historical sub-PSF approach, and mapping out the extended capture range capabilities. Finally, Sec. 5 summarizes our core findings and discusses paths toward future experimental validation.

2. SIMULATION FRAMEWORK

We describe in this section the simulation and machine learning pipelines used to generate and learn from the data. As stated in the introduction, this work is an evolution of the method proposed earlier in Ref. 3; hence, we omit an exhaustive description of the simulation environment. We instead summarize the fundamental technical components, inviting the reader to review our previous publications for further architectural details.

To that end, Sec. 2.1 provides a brief formulation of the differential piston metrics, while Sec. 2.2 describes the introduction of phase diversity to break the even-mode degeneracy inside the pupil plane. We then define the performance estimators utilized to quantify our regression accuracy in Sec. 2.3, followed by an outline of the fundamental optical and atmospheric parameters used in our simulations in Sec. 2.4. Finally, the neural network architecture is presented in Sec. 2.5, and the corresponding dataset generation, sample size, and data-splitting protocols are detailed in Sec. 2.6.

2.1 Petal definition

We simulate an ELT-like pupil with 6 petals that undergo piston aberrations using the P3 library. The petals are labeled from 1 to 6 and their respective pistons are denoted p_i with $i \in [1, 6]$. Hence, we define the differential piston between petals i and j as the difference in their respective piston values p_i and p_j , yielding:

$$\Delta p_{\{j,i\}} = p_j - p_i \quad \text{with} \quad i, j \in [1, 6], \quad i \neq j. \quad (1)$$

In this paper, we consider different amplitudes for the input piston error. In Sec. 3.1, we limit the error to the classical capture range of $[-\pi, \pi]$, but in Sec. 3.3 we see how our method is able to extend to higher amplitudes up to a range of $[-6\pi, 6\pi]$.

The shape of the distribution is chosen to be a uniform distribution over the individual petal pistons p_i :

$$p_i \sim \mathcal{U}(-\pi/2, \pi/2) \quad \text{with} \quad i \in [1, 6], \quad (2)$$

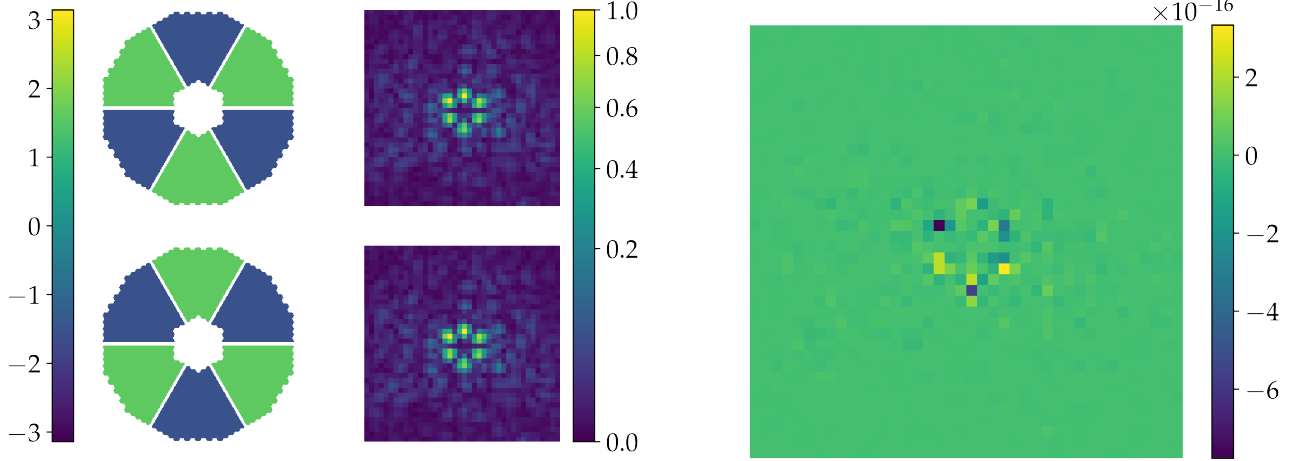


Figure 1: Illustration of the symmetry obtained in the Fourier domain for two even modes (top and bottom). Left: Phase in the pupil plane. Center: Normalized PSF intensity in the focal plane. Right: Difference between top and bottom PSF showing values close to computing precision.

which leads to a triangular distribution over the differential pistons $\Delta p_{j,i}$:

$$\Delta p_{j,i} \sim \mathcal{T}(-\pi, \pi) \quad \text{with} \quad i, j \in [1, 6], \quad i \neq j. \quad (3)$$

2.2 Breaking the symmetry

We know from Ref. 5 that even phase modes cannot be sensed in the absence of asymmetry in the telescope pupil. Based on a LIFT-like configuration,⁶ where a phase diversity is added to the pupil, we choose to introduce a defocus of 1 radian RMS at the central wavelength of $\lambda_0 = 1.65 \mu\text{m}$ to the pupil in order to break the degeneracy. From a practical point of view, this defocus can be obtained either by adding a phase inside the pupil plane or by moving the detector away from the focal point.

Figure 1 illustrates the arising degeneracy in the PSF plane. For even petal modes, the resulting PSF is identical; therefore, we lose the sign of the mode and cannot correct for it.

2.3 Performance estimators

The optical performance of our method is evaluated using the total root mean squared error (RMSE) over the 5 differential pistons, defined as:

$$\text{RMSE} = \sqrt{\frac{1}{5} \sum_{i=1}^5 \text{RMSE}_{\{i+1,i\}}^2}, \quad (4)$$

with:

$$\text{RMSE}_{\{j,i\}} = \sqrt{\frac{1}{N} \sum_{k=1}^N \left(\widehat{\Delta p_{\{j,i\}}^k} - \Delta p_{\{j,i\}}^k \right)^2}, \quad (5)$$

where $\widehat{\Delta p_{\{j,i\}}^k}$ is the estimated value of the true differential piston $\Delta p_{\{j,i\}}^k$ for the k -th sample.

2.4 Optical parameters

In this paper, no detector or photon noise has been added to the images, and a single atmospheric configuration has been evaluated: a 25% Strehl ratio baseline with a 13-frame temporal average over uncorrelated phase screens. These core simulation and structural parameters are summarized in Tab. 1.

Table 1: List and main characteristics of the datasets used in our simulations.

Pupil	ELT pupil ^(a) Monolithic primary with 50 cm spiders Central obstruction 13.5 m Resolution: 238 px
Piston	Uniform distribution on the 6 petals $\alpha \in [-\pi, \pi]$ with $\alpha \in \mathbb{R}^{+*}$
Turbulence	From TIPTOP ^(b) PSD Independent phasescreen Strehl ratio at 25%
PSF	Two monochromatic channels : $\lambda_0 - \Delta\lambda/2$ and $\lambda_0 + \Delta\lambda/2$ H-band-like : $\lambda_0 = 1.65\mu\text{m}$ and $\Delta\lambda = 0.3\mu\text{m}$ Resolution: 96 px Sampling: 2 px per λ/D Defocus: 1 rad

^(a) from P3 library (Ref. 7), ^(b)Ref. 8

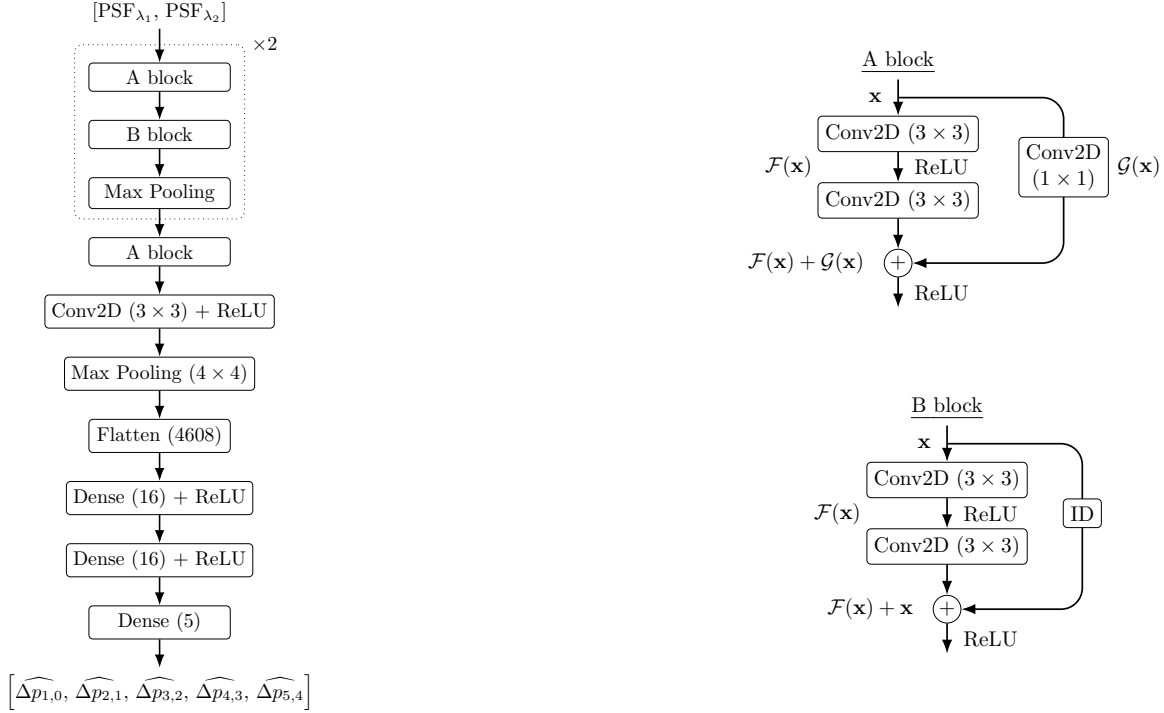


Figure 2: Left: ResNet architecture. The number of filters is shown to the right of each convolutional layer. Right: Residual blocks used in the ResNet. Top: A-block type, where the skip layer is a 1×1 convolution. Bottom: B-block type, where the skip layer is the identity mapping.

2.5 Machine learning pipeline

In contrast to the method presented in Ref. 3, our current objective is to estimate the five differential pistons concurrently from a single defocused image. Consequently, two key modifications were made to the network topology: first, the input layer was adapted to accept a two-color PSF (utilized for the capture range extension detailed in Sec. 3.3); second, the output layer was modified to exclude the pupil rotation angle metric used in

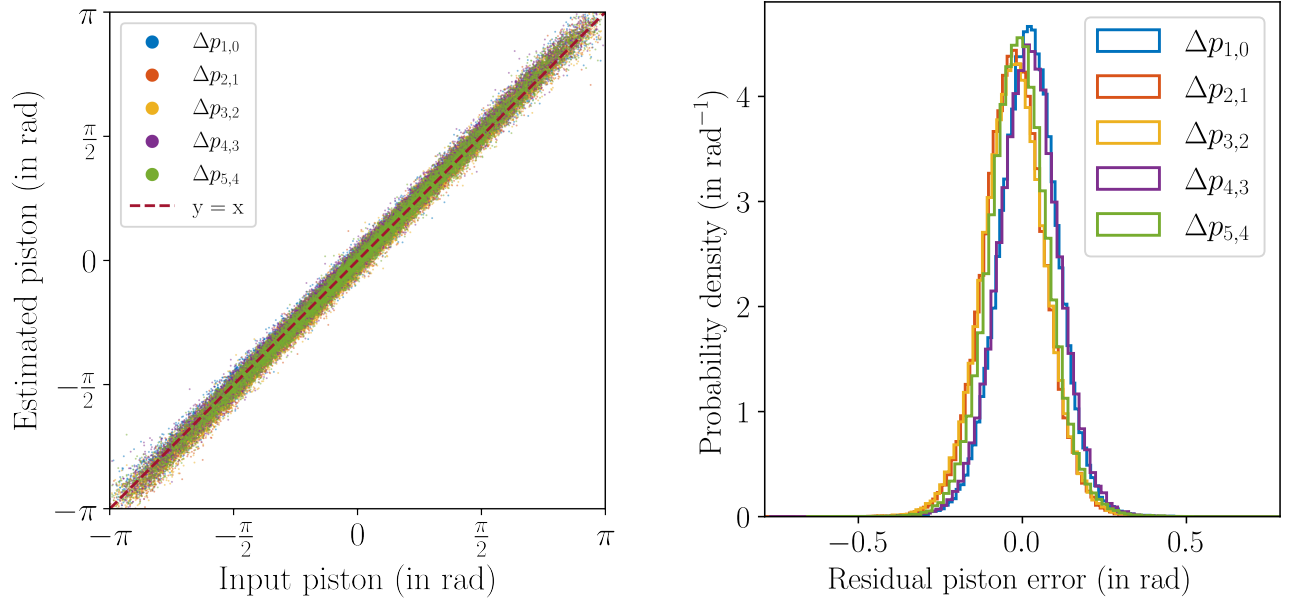


Figure 3: Left: Estimated differential piston as a function of the introduced differential piston. Right: Probability density of the residual piston error. The five differential piston pairs are color-coded to highlight their respective contributions.

the previous iteration. This structural simplification is supported by our assumption that the PSF field does not undergo rotation.

The network schematic is illustrated in Fig. 2. The training configuration closely follows the optimization protocols described in Refs. 3, 4, utilizing batches of 128 elements to minimize the Huber loss of the prediction residuals.

2.6 Datasets

We generate 240 000 defocused PSFs, which are split into 100 000 PSFs for training, 50 000 PSFs for validation, and 90 000 PSFs for testing. The training samples are strictly reserved for the optimization loop; therefore, the network never encounters images belonging to the validation or test sets during training. Compared to our previous study, this setup represents a one-third reduction in the number of training images, significantly accelerating the optimization process. This reduction is justified by our assumption of a fixed pupil orientation; eliminating pupil rotation reduces the overall diversity of the dataset, allowing the network to converge with equal efficiency on a smaller sample size.

3. RESULTS

In this section, we present the performance evaluations of our proposed method. First, Sec. 3.1 analyzes the regression accuracy within the classical $[-\pi, \pi]$ capture range baseline. Next, Sec. 3.2 benchmarks these results against the performance of the historical sub-PSF approach described in Ref. 3. Finally, Sec. 3.3 demonstrates the expanded capture range thresholds achieved by exploiting dual-wavelength diversity.

3.1 Performance in the $[-\pi, \pi]$ range

Figure 3 (left) displays the scatter plot of the estimated differential pistons as a function of their introduced values across all five segment pairs. As expected, the network outputs exhibit a robust linear relationship with well-bounded residuals. Figure 3 (right) presents the probability density functions of the residual errors (i.e., the difference between the predicted values and the $y = x$ identity line). The five distributions are remarkably similar, save for slight variations in their respective means. These minor offsets are attributed to the preliminary nature

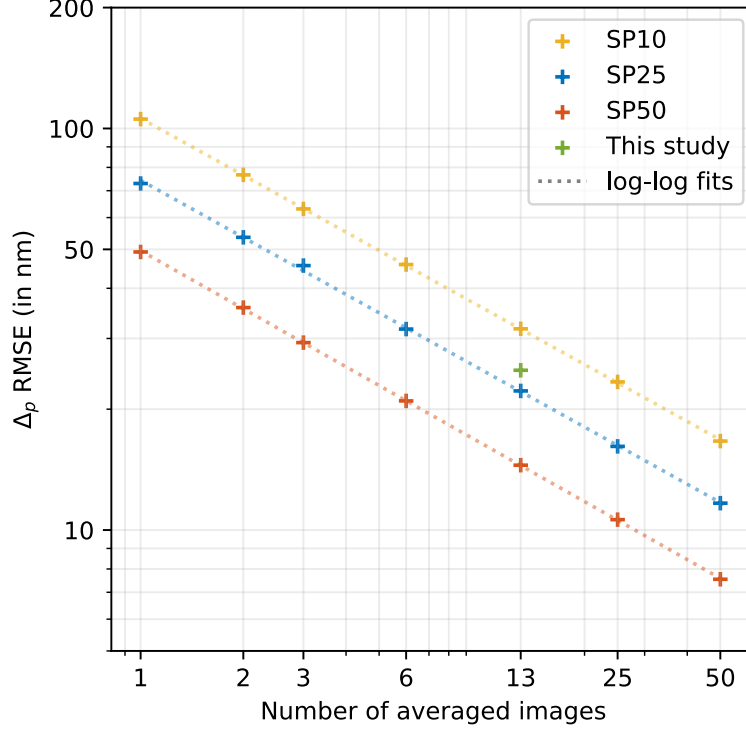


Figure 4: Differential piston RMSE as a function of the number of averaged images extracted from Ref. 3, with the result from this study added as a green cross.

of the network optimization, as these initial results were obtained using fixed, untuned hyperparameters over a brief training run of only 30 epochs, and are expected to vanish with a optimized hyperparameter configuration and extended training.

Quantifying these errors using the metrics defined in Sec. 2.3 yields:

$$\begin{aligned}
 \text{RMSE}_{\{2,1\}} &= 0.0896 \text{ rad}, \\
 \text{RMSE}_{\{3,2\}} &= 0.0941 \text{ rad}, \\
 \text{RMSE}_{\{4,3\}} &= 0.0966 \text{ rad}, \\
 \text{RMSE}_{\{5,4\}} &= 0.0935 \text{ rad}, \\
 \text{RMSE}_{\{6,5\}} &= 0.0925 \text{ rad},
 \end{aligned}$$

resulting in a global wavefront error of $\text{RMSE} = 0.0933 \text{ rad}$, which translates to approximately 25 nm RMS at the central wavelength of $\lambda_0 = 1.65 \mu\text{m}$.

3.2 Comparison with the sub-PSF approach

Comparing these to the results obtained in Ref. 3 under the same physical conditions, we observe a very close agreement, with a 23 nm RMSE in the previous study versus a 25 nm RMSE in this work. Figure 4 presents this new value added as a green cross on the plot adapted from Ref. 3. The specific configuration evaluated in this work corresponds to the 25% Strehl ratio case (SP25, represented by the blue curve) with a 13-frame average.

Note that in the current study, we use a preliminary setup with fixed hyperparameters and only 30 epochs of learning. Consequently, this 2 nm difference is not highly significant, and we can consider that the system reaches an identical level of performance. This is a promising result since the network now estimates all 5 differential pistons simultaneously, compared to the single differential piston estimation in the previous case. Moreover, the current method is robust against pupil rotation, whereas the previous approach exhibited performance degradation at pupil rotations close to 0° or 30° .

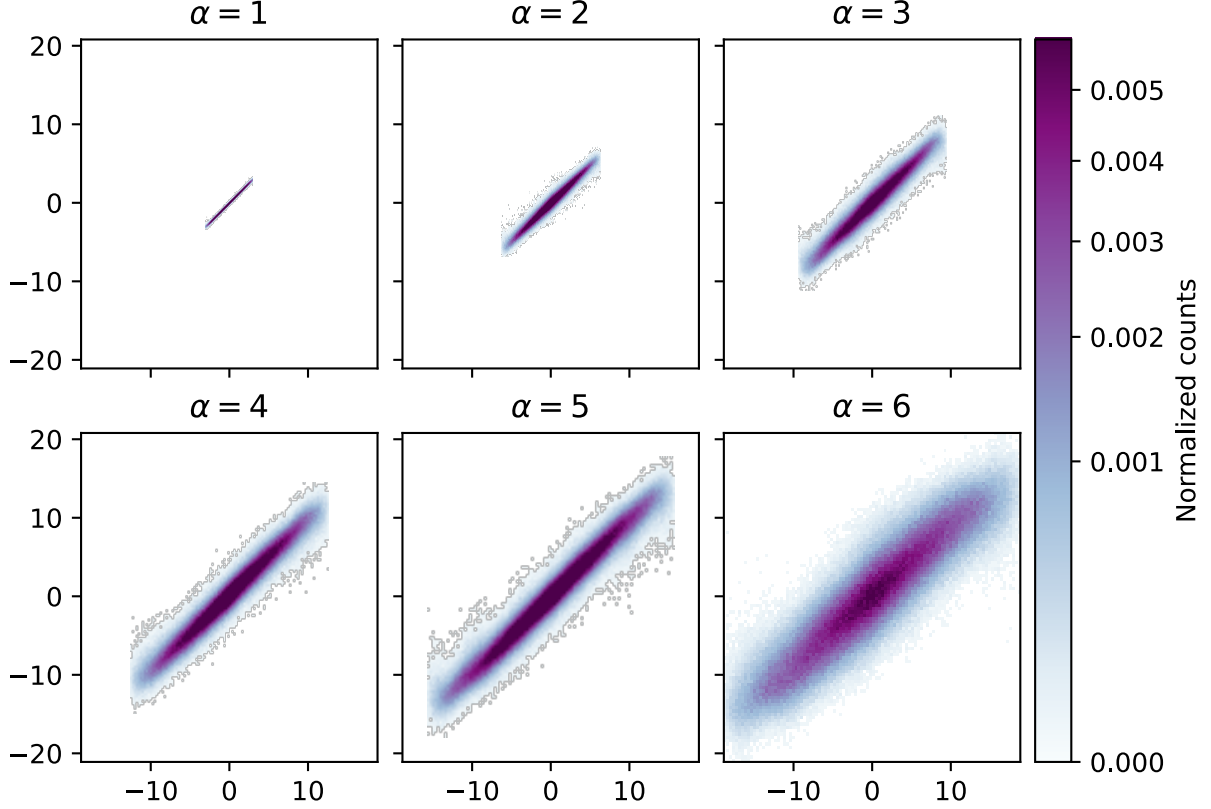


Figure 5: 2D histograms of the introduced versus estimated piston values for different capture range expansion factors (α). The value of α is indicated at the top of each panel, and a consistent color scale is applied across all subplots for direct comparison.

3.3 Enhancement of the capture range

In this section, we evaluate the expansion of the piston capture range enabled by dual-wavelength input. To this end, we allow the initial differential piston values to exceed the classical $[-\pi, \pi]$ capture range. Specifically, we train the neural network across five expanded capture ranges defined by $\alpha[-\pi, \pi]$, where $\alpha \in \{2, 3, 4, 5, 6\}$.

It is well-established that combining two or more distinct wavelengths can resolve the phase ambiguity and extend the effective capture range up to a certain threshold dictated by the choice of these spectral configurations.^{3,9–13} Here, we implement the straightforward case of a dual-wavelength input configuration. This setup theoretically extends the phase measurement baseline to $[-\Lambda/2, \Lambda/2]$, where Λ represents the synthetic (or beat) wavelength defined as:

$$\Lambda = \frac{\lambda_1 \lambda_2}{\lambda_2 - \lambda_1} \quad \text{with} \quad \lambda_2 > \lambda_1. \quad (6)$$

Using our specific spectral parameters ($\lambda_1 = 1.50 \mu\text{m}$ and $\lambda_2 = 1.80 \mu\text{m}$), the resulting synthetic wavelength is $\Lambda = 9.0 \mu\text{m} \simeq 5.45 \lambda_0$.

Following the methodology outlined in Sec. 2.6, we trained a dedicated neural network for each value of α and evaluated their performance using the 90,000 PSFs from the independent test set.

Figure 5 displays the 2D histograms of the introduced versus estimated piston values across all evaluated values of α , while Fig. 6 displays the total residual RMSE (Eq. 4) against the standard deviation of the input differential pistons. For ease of reference, the corresponding input capture range boundaries are indicated along the top horizontal axis of the latter plot.

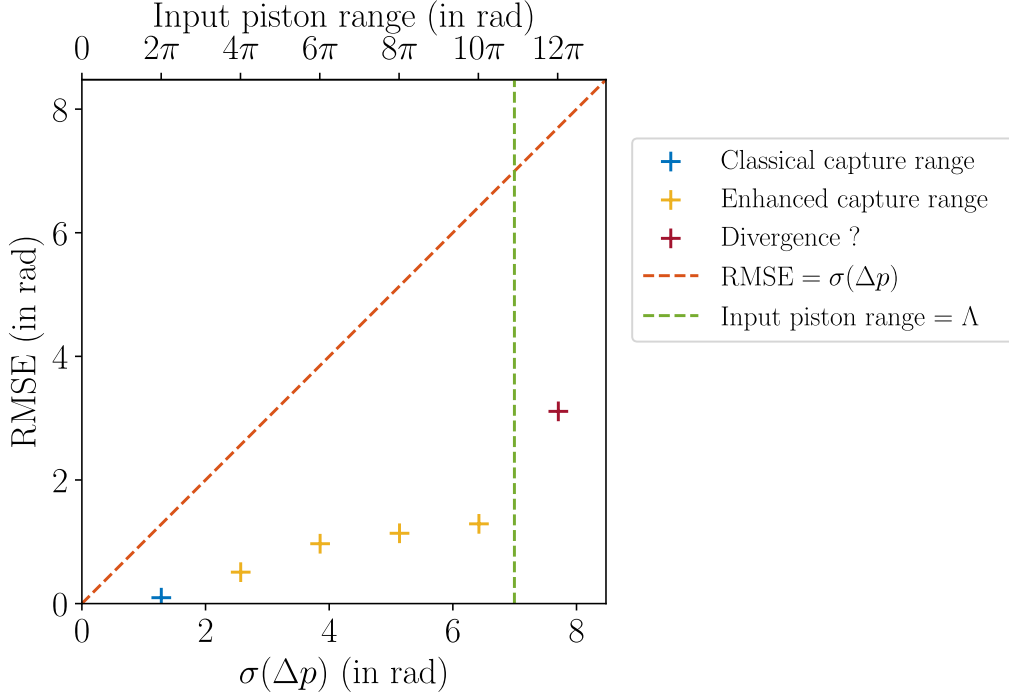


Figure 6: Total residual RMSE (Eq. 4) as a function of the standard deviation of the input differential pistons. The blue cross denotes the baseline $[-\pi, \pi]$ capture range, the yellow crosses represent the expanded capture ranges up to the synthetic wavelength limit $[-\Lambda, \Lambda]$, and the dark red cross marks the regime exceeding this theoretical dual-wavelength threshold. For ease of reference, the corresponding input capture range boundaries are indicated along the top horizontal axis.

As expected, the 2D histograms in Fig. 5 closely follow the ideal linear identity trend, with the variance around the diagonal visibly increasing alongside the value of α . This progression is quantitatively confirmed by the global performance curve in Fig. 6.

These results demonstrate that our models are capable of reliably estimating differential pistons well beyond the classical baseline capture range. For values of α up to 5, the overall dispersion of the datapoints remains qualitatively similar, whereas a distinct change in regime becomes apparent at $\alpha = 6$. This behavioral shift is explicitly mirrored in Fig. 6, where the dark red datapoint ($\alpha = 6$) exhibits a sharp departure in the output versus input RMSE trend.

This behavior is expected when considering Eq. 6, as $\alpha = 6$ corresponds to a capture range that exceeds the synthetic wavelength Λ . However, inspecting Fig. 5 reveals that the system still maintains a coarse linear behavior, even though the variance around the ideal diagonal response is proportionally larger than for lower values of α . Under these conditions, the classic manifestation of phase wrapping would typically be a sign uncertainty near the boundaries of the capture range (i.e., data points being assigned an incorrect sign for the piston value); however, this does not appear to occur here. A more detailed examination of this behavior remains a subject for future work.

Crucially, while these open-loop residual errors appear non-trivial at high amplitudes, they represent a single-shot estimation. When integrated into a closed-loop adaptive optics control sequence, these residuals are expected to converge rapidly. Based on these open-loop trends, we are confident that the system can reliably close the loop across the entire theoretical $[-\Lambda/2, \Lambda/2]$ window, ultimately bringing the final residual pistons down to the sub-radian regime achieved in our baseline $[-\pi, \pi]$ trials.

4. FUTURE EXPERIMENTAL WORK

One of our primary, long-term objectives is to demonstrate the practical viability of this deep-learning framework under real-world, on-sky conditions. To bridge the gap between numerical simulations and operational telescope environments, we will initiate a systematic experimental validation campaign using data obtained from a dedicated laboratory optical bench. Specifically, initial testing will be conducted on the High-Contrast Module (HCM) development testbed¹⁴ hosted at IPAG. This bench is uniquely suited for our validation needs as it features an integrated Zernike wavefront sensor (Z-WFS) alongside the optical infrastructure required to record defocused focal-plane images. In this configuration, the ZELDA sensor will serve as a baseline reference, allowing us to directly compare the network’s prediction accuracy against a standard, established wavefront sensing method.

Our experimental roadmap is structured into three sequential phases:

1. **Direct transfer:** The first step will evaluate the performance of our existing, simulation-trained networks directly on the raw bench data without any structural modifications. This will allow us to quantify the baseline domain gap between our analytical Fourier models and real physical hardware.
2. **Robustness via data augmentation:** If the zero-shot transfer fails to meet our performance thresholds, we anticipate that the primary source of discrepancy will stem from optical mismatches, particularly variations in the nominal defocus amplitude between the simulation dataset and the physical bench setup. To overcome this, we will retrain the models using augmented datasets that incorporate a randomized variety of defocus levels, explicitly teaching the network to remain invariant to small structural alignment tolerances.
3. **On-sky deployment:** Once the framework is experimentally validated and its robustness is confirmed on the laboratory bench, we aim to transition the system to a night-sky environment. The ultimate milestone will involve deploying the optimized pipeline for an on-sky demonstration, targeting a technical run on the EKARUS instrument.¹⁵ As an adaptive optics R&D platform located at the Asiago Observatory, EKARUS features a dedicated pupil fragmentation unit that provides the capability to precisely introduce, modulate, and control differential piston aberrations in a real-world observing environment.

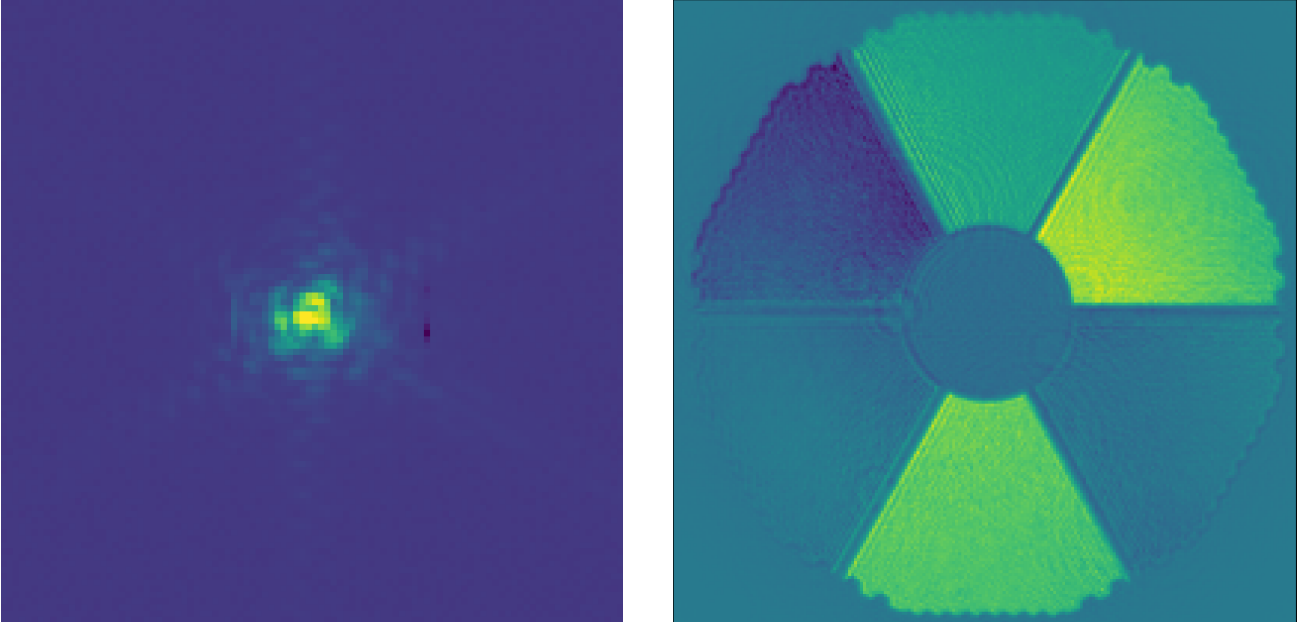


Figure 7: Left: Defocused PSF obtained on the HCM bench at IPAG. Right: Corresponding Zernike-WFS phase measurement.

5. CONCLUSION

In this paper, we have presented an evolution of our deep-learning framework dedicated to estimating differential piston aberrations, which represent a critical limitation for the optical performance of the ELT. By replacing our previous sub-PSF approach with a method exploiting defocused dual-wavelength focal-plane images, we successfully enhanced both (1) the capture range performance and (2) the overall versatility of the sensor.

Regarding the first advancement, the network can concurrently estimate all five differential piston modes of a 6-petal pupil from a single image frame. Under realistic ELT turbulence residuals (25% Strehl ratio baseline with 13-frame averaging), the network achieves a wavefront error of approximately 25 nm RMS. This matches the performance of the historical framework while introducing native robustness against pupil rotation.

Regarding the second advancement, by incorporating a dual-wavelength input configuration, we demonstrated that the network effectively expands its operational bounds well beyond the classical $[-\pi, \pi]$ radian phase ambiguity. Robust linear estimation is maintained up to an expansion factor of $\alpha = 5$. At $\alpha = 6$, where the physical capture range surpasses the theoretical synthetic wavelength limit of $\Lambda = 9.0 \mu\text{m}$, a distinct change in the performance regime occurs; remarkably, however, the system preserves a coarse linear behavior without clear signs of classical edge-boundary phase wrapping.

While these open-loop estimations exhibit increased variance at large initial piston amplitudes, their performance is more than sufficient to initialize a correction loop. Embedded within a closed-loop control sequence, these single-shot estimations are expected to drive the system into rapid convergence, systematically reducing massive phase errors down to the sub-radian regime.

A more detailed examination of this closed-loop control scheme, alongside the specific behavioral regime encountered beyond the synthetic wavelength boundary, remains a subject for future work. Moving forward, our next steps will focus on hyperparameter optimization to eliminate the minor systematic offsets noted during brief training runs, followed by experimental validation on dedicated optical testbeds and on-sky to confirm the framework's viability for upcoming ELT instruments such as HARMONI.

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